

Trade Nexus - Complete Technical Documentation

Version: 1.0 **Date:** October 29, 2025 **Application:** Trade Nexus - Trading Risk Management & Compliance Platform

Table of Contents

- 1. [Executive Summary](#)
- 2. [System Architecture](#)
- 3. [Technology Stack](#)
- 4. [API Endpoints](#)
- 5. [Application Routes & Pages](#)
- 6. [Services & Business Logic](#)
- 7. [User Roles & Permissions](#)
- 8. [Authentication & Security](#)
- 9. [Database Schema](#)
- 10. [Data Flow & Processing](#)
- 11. [Frontend Components](#)
- 12. [Deployment & Infrastructure](#)

Executive Summary

Trade Nexus is a comprehensive, enterprise-grade trading risk management and compliance platform designed for commodity trading operations. The system provides real-time position monitoring, regulatory compliance reporting, pre-trade validation, and multi-tenant architecture supporting exchanges, companies, and traders.

Key Features

- **50+ API Endpoints** covering all trading operations
- **36 Frontend Pages** providing complete UI coverage
- **35 Service Modules** implementing complex business logic
- **70+ Database Tables** with comprehensive audit trails
- **6-Tier Role-Based Access Control** with 100+ granular permissions
- **Multi-Tenancy Architecture** (Exchange → Company → Trader)
- **Real-Time Position Monitoring** with breach detection
- **Regulatory Reporting** for CFTC, ICE, and CME
- **SSO/SAML Integration** with 2FA support
- **Document Management** with versioning
- **Support Ticket System** and communication tools

Platform Statistics

Category	Count
API Endpoints	50+
Frontend Pages	36

Business Services	35
Database Tables	70+
User Roles	6
Permissions	100+
Database Migrations	27
Lines of Service Code	~12,500

System Architecture

Architecture Overview

Trade Nexus follows a modern, cloud-native architecture deployed entirely on Cloudflare's edge network:

Frontend (Cloudflare Pages)

- Next.js 15.1.3 with App Router
- React 18.3.1
- Tailwind CSS + shadcn/ui components
- Zustand for state management
- Recharts for data visualization

Backend (Cloudflare Workers)

- Hono framework (v4.6.14) - lightweight web framework
- TypeScript for type safety
- Middleware-based request processing
- RESTful API design

Data Layer (Cloudflare)

- D1 Database (SQLite) - primary data store
- KV Store - caching and session management
- R2 Storage - document and file storage

External Integrations

- Exchange APIs (ICE, CME, CFTC)
- SAML/OAuth SSO providers
- Email service providers

Deployment Model

- **Edge Computing:** Deployed globally on Cloudflare's edge network
 - **Low Latency:** Sub-100ms response times globally
 - **High Availability:** 99.99% uptime SLA
 - **Auto-Scaling:** Automatic scaling based on demand
 - **Zero Downtime Deployments:** Rolling updates with instant rollback
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Technology Stack

Backend Technologies

Component	Technology	Version
Runtime	Cloudflare Workers	Latest
Framework	Hono	4.6.14
Language	TypeScript	5.x
Database	Cloudflare D1 (SQLite)	Latest
Cache	Cloudflare KV	Latest
Storage	Cloudflare R2	Latest
Validation	Zod	Latest
Authentication	JWT	Latest

Frontend Technologies

Component	Technology	Version
Framework	Next.js	15.1.3
UI Library	React	18.3.1
Language	TypeScript	5.x
Styling	Tailwind CSS	3.x
Components	shadcn/ui	Latest
State	Zustand	5.0.1
Charts	Recharts	2.12.7
Forms	React Hook Form	Latest

Development Tools

- **Build Tool:** Vite
- **Testing:** Vitest + Testing Library
- **Linting:** ESLint
- **Formatting:** Prettier
- **Version Control:** Git
- **CI/CD:** GitHub Actions / Cloudflare Pages

API Endpoints

Complete API Reference

1. Authentication & User Management

Base Path: /api/auth

Endpoint	Method	Description	Auth Required
/register	POST	Register new user	No
/login	POST	User login with JWT	No
/refresh	POST	Refresh access token	Yes
/logout	POST	User logout	Yes
/me	GET	Get current user info	Yes
/forgot-password	POST	Request password reset	No
/reset-password	POST	Reset password with token	No
/detect-auth-method	POST	Detect SSO/SAML config	No

SAML SSO (/api/auth/saml)

Endpoint	Method	Description	Auth Required
/login	GET	Initiate SAML login	No
/callback	POST	SAML assertion callback	No
/logout	POST	SAML logout	Yes
/metadata/:companyId	GET	Company SAML metadata	No

OAuth (/api/auth/oauth)

Endpoint	Method	Description	Auth Required
/login	GET	Initiate OAuth flow	No
/callback	GET	OAuth callback	No
/refresh	POST	Refresh OAuth token	Yes
/logout	POST	OAuth logout	Yes

Users (/api/users)

Endpoint	Method	Description	Auth Required
/profile	GET	Get user profile	Yes
/profile	PUT	Update user profile	Yes
/settings	GET	Get user settings	Yes
/settings	PUT	Update user settings	Yes

/change-password	POST	Change password	Yes
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2. Dashboard & Monitoring

Dashboard (/api/dashboard)

Endpoint	Method	Description	Auth Required
/overview	GET	Real-time dashboard overview	Yes
/by-commodity	GET	Position data by commodity	Yes
/trending	GET	Trending positions and risks	Yes
/heatmap	GET	Position utilization heatmap	Yes
/concentration	GET	Position concentration analysis	Yes
/commodity/:code	GET	Commodity-specific data	Yes
/ice-positions	GET	ICE exchange positions	Yes

Monitoring (/api/monitoring)

Endpoint	Method	Description	Auth Required
/history	GET	Historical monitoring data	Yes
/breaches	GET	Position limit breaches	Yes
/breaches/:id	GET	Breach details	Yes
/breaches/:id	PATCH	Update breach status	Yes
/run	POST	Trigger manual monitoring	Yes
/compliance/audit	GET	Compliance audit trail	Yes
/data-quality	GET	Data quality metrics	Yes
/stats	GET	Monitoring statistics	Yes

Performance (/api/performance)

Endpoint	Method	Description	Auth Required
/stats	GET	Request performance stats	Yes
/rate-limits	GET	Rate limiting statistics	Yes

3. Position & Limit Management

Position Limits (/api/position-limits)

Endpoint	Method	Description	Auth Required
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/	GET	List position limits	Yes
/:id	GET	Get position limit details	Yes
/	POST	Create position limit	Yes
/:id	PUT	Update position limit	Yes
/:id	DELETE	Delete position limit	Yes

Market Limits (/api/market-limits)

Endpoint	Method	Description	Auth Required
/	GET	List market limits	Yes
/:id	GET	Get market limit by ID	Yes
/commodity/:code	GET	Get limits by commodity	Yes
/	POST	Create market limit	Yes
/:id	PUT	Update market limit	Yes
/:id	DELETE	Delete market limit	Yes

Transactions (/api/transactions)

Endpoint	Method	Description	Auth Required
/	GET	List transactions	Yes
/:id	GET	Get transaction details	Yes
/	POST	Create transaction	Yes
/:id	DELETE	Delete transaction	Yes

4. Risk Management

Pre-Trade (/api/pre-trade)

Endpoint	Method	Description	Auth Required
/validate	POST	Validate single trade	Yes
/batch-validate	POST	Batch trade validation	Yes
/history	GET	Pre-trade check history	Yes
/stats	GET	Pre-trade statistics	Yes

Risk Thresholds (/api/risk-thresholds)

Endpoint	Method	Description	Auth Required
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/	GET	List risk thresholds	Yes
/:id	GET	Get threshold details	Yes
/	POST	Create threshold	Yes
/:id	PUT	Update threshold	Yes
/:id	DELETE	Delete threshold	Yes

Risk Scenarios (/api/risk-scenarios)

Endpoint	Method	Description	Auth Required
/	GET	List scenarios	Yes
/:id	GET	Get scenario details	Yes
/	POST	Create scenario	Yes
/:id	PUT	Update scenario	Yes
/:id	DELETE	Delete scenario	Yes
/:id/run	POST	Execute scenario	Yes
/:id/results	GET	Get scenario results	Yes
/run-all	POST	Execute all scenarios	Yes

Risk Metrics (/api/risk-metrics)

Endpoint	Method	Description	Auth Required
/	GET	Get risk metrics	Yes
/var	GET	Value at Risk calculations	Yes
/stress-tests	GET	Stress test results	Yes

5. Compliance & Regulatory

Exemptions (/api/exemptions)

Endpoint	Method	Description	Auth Required
/	GET	List exemptions	Yes
/stats	GET	Exemption statistics	Yes
/:id	GET	Get exemption details	Yes
/	POST	Create exemption	Yes
/:id/approve	POST	Approve exemption	Yes
/:id/deny	POST	Deny exemption	Yes

/:id	DELETE	Delete exemption	Yes
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Regulatory Filings (/api/regulatory-filings)

Endpoint	Method	Description	Auth Required
/	GET	List filings	Yes
/:id	GET	Get filing details	Yes
/generate/cftc-ltrs	POST	Generate CFTC LTRS report	Yes
/generate/ice-daily	POST	Generate ICE daily report	Yes
/generate/cme	POST	Generate CME report	Yes
/:id/submit	POST	Submit filing	Yes
/:id/download	GET	Download filing	Yes
/stats/summary	GET	Filing statistics	Yes

Bid Week (/api/bid-week)

Endpoint	Method	Description	Auth Required
/holidays	GET	Get exchange holidays	Yes
/holidays	POST	Create holiday	Yes
/holidays/:exchange/:date	PUT	Update holiday	Yes
/holidays/:exchange/:date	DELETE	Delete holiday	Yes
/holidays/bulk-import	POST	Bulk import holidays	Yes
/schedules	GET	Get bid week schedules	Yes
/schedules/generate	POST	Generate schedules	Yes
/schedules/regenerate	POST	Regenerate schedules	Yes
/status	GET	Current bid week status	Yes
/status/all	GET	All bid week statuses	Yes
/spot-month	GET	Spot month information	Yes
/auto-generate	POST	Auto-generate schedules	Yes
/gbd-check	GET	Gas Business Day check	Yes

6. Alerts & Notifications

Alerts (/api/alerts)

Endpoint	Method	Description	Auth Required
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/	GET	List alerts	Yes
/stats	GET	Alert statistics	Yes
/:id	GET	Get alert details	Yes
/	POST	Create alert	Yes
/:id/read	PATCH	Mark as read	Yes
/:id/acknowledge	POST	Acknowledge alert	Yes
/:id	DELETE	Delete alert	Yes
/rules	GET	Get alert rules	Yes
/rules/:id	PUT	Update alert rule	Yes

7. Reports

Reports (/api/reports)

Endpoint	Method	Description	Auth Required
/position	GET	Position report	Yes
/compliance	GET	Compliance report	Yes
/breaches	GET	Breaches report	Yes
/historical	GET	Historical report	Yes
/pre-trade	GET	Pre-trade report	Yes
/approvals	GET	Approvals report	Yes
/audit	GET	Audit report	Yes
/export	POST	Export report (CSV/Excel)	Yes
/summary	GET	Report summary	Yes

Subset Reports (/api/subset-reports)

Endpoint	Method	Description	Auth Required
/top-counterparties	GET	Top counterparties report	Yes
/next-day-fixed	GET	Next day fixed price report	Yes
/next-day-index	GET	Next day index report	Yes
/next-day-exposure	GET	Next day exposure report	Yes
/generate	POST	Generate subset report	Yes
/:id	GET	Get report details	Yes

/	GET	List reports	Yes
/cleanup	DELETE	Cleanup old reports	Yes

8. Data Management

Aggregation (/api/aggregation)

Endpoint	Method	Description	Auth Required
/groups	GET	Get aggregation groups	Yes
/groups/:id/members	GET	Get group members	Yes
/positions	GET	Aggregated positions	Yes
/calculate	POST	Calculate aggregation	Yes
/equivalence	POST	Calculate equivalence	Yes
/spread-netting	POST	Calculate spread netting	Yes
/relationships/:commodity	GET	Commodity relationships	Yes
/stats	GET	Aggregation statistics	Yes

Data Quality (/api/data-quality)

Endpoint	Method	Description	Auth Required
/dashboard	GET	Data quality dashboard	Yes
/issues	GET	Data quality issues	Yes
/rules	GET	Quality rules	Yes
/rules	POST	Create rule	Yes
/rules/:id	PUT	Update rule	Yes
/rules/:id	DELETE	Delete rule	Yes
/run	POST	Run quality checks	Yes
/reconciliation	GET	Reconciliation report	Yes
/reconciliation	POST	Create reconciliation	Yes
/lineage/:table/:id	GET	Data lineage	Yes
/issues/:id	PUT	Update issue	Yes
/uploads	GET	Upload history	Yes
/stats	GET	Quality statistics	Yes

Mapping (/api/mapping)

Endpoint	Method	Description	Auth Required
/	GET	List mappings	Yes
/:id	GET	Get mapping	Yes
/	POST	Create mapping	Yes
/:id	PUT	Update mapping	Yes
/:id	DELETE	Delete mapping	Yes
/commodity/:code	GET	Mappings by commodity	Yes
/market/:location	GET	Mappings by market	Yes

CSV Import (/api/csv-import)

Endpoint	Method	Description	Auth Required
/transactions	POST	Import transaction CSV	Yes
/power-data	POST	Import power data CSV	Yes

9. Admin & Configuration

Companies (/api/companies)

Endpoint	Method	Description	Auth Required
/	GET	List companies	Yes
/:id	GET	Get company details	Yes
/	POST	Create company	Yes (Admin)
/:id	PUT	Update company	Yes (Admin)
/:id/traders	GET	Company traders	Yes
/:id/traders	POST	Add trader to company	Yes

Traders (/api/traders)

Endpoint	Method	Description	Auth Required
/	GET	List traders	Yes
/:id	GET	Get trader details	Yes
/	POST	Create trader	Yes
/:id	PUT	Update trader	Yes
/:id	DELETE	Delete trader	Yes

Exchanges (/api/exchanges)

Endpoint	Method	Description	Auth Required
/	GET	List exchanges	Yes
/:id	GET	Get exchange details	Yes
/	POST	Create exchange	Yes (Super Admin)
/:id	PUT	Update exchange	Yes (Super Admin)
/:id	DELETE	Delete exchange	Yes (Super Admin)

API Keys (/api/api-keys)

Endpoint	Method	Description	Auth Required
/	GET	List API keys	Yes
/	POST	Create API key	Yes
/:id	GET	Get key details	Yes
/:id/usage	GET	Key usage statistics	Yes
/:id	PUT	Update key	Yes
/:id	DELETE	Revoke key	Yes
/stats/summary	GET	API key summary	Yes

10. Workflow & Approvals

Approvals (/api/approvals)

Endpoint	Method	Description	Auth Required
/	GET	List approval requests	Yes
/	POST	Create approval request	Yes
/:id/approve	POST	Approve request	Yes
/:id/reject	POST	Reject request	Yes
/:id/history	GET	Approval history	Yes

Trade Approvals (/api/trade-approvals)

Endpoint	Method	Description	Auth Required
/	GET	List trade approvals	Yes
/pending-count	GET	Count pending approvals	Yes
/	POST	Submit for approval	Yes
/:id/approve	POST	Approve trade	Yes

/:id/reject	POST	Reject trade	Yes
/history	GET	Approval history	Yes

11. Documents & Support

Documents (/api/documents)

Endpoint	Method	Description	Auth Required
/	GET	List documents	Yes
/	POST	Upload document	Yes
/:id	GET	Get document details	Yes
/:id/download	GET	Download document	Yes
/:id/versions	GET	Document versions	Yes
/:id/approve	POST	Approve document	Yes
/:id/archive	POST	Archive document	Yes
/recent	GET	Recent documents	Yes
/expiring	GET	Expiring documents	Yes

Support (/api/support)

Endpoint	Method	Description	Auth Required
/tickets	GET	List support tickets	Yes
/tickets	POST	Create ticket	Yes
/tickets/:id/messages	GET	Ticket messages	Yes
/tickets/:id/messages	POST	Add message	Yes
/tickets/:id/status	PATCH	Update ticket status	Yes
/newsletters	GET	List newsletters	Yes
/newsletters	POST	Create newsletter	Yes (Admin)
/newsletters/:id/publish	POST	Publish newsletter	Yes (Admin)
/newsletters/:id/read	POST	Mark as read	Yes

12. Audit & Security

Audit Trail (/api/audit-trail)

Endpoint	Method	Description	Auth Required
/	GET	Query audit entries	Yes

/stats	GET	Audit statistics	Yes
/entity/:type/:id	GET	Entity history	Yes
/user/:userId	GET	User activity	Yes
/search	GET	Search audit trail	Yes
/	POST	Manual audit entry	Yes
/cleanup	DELETE	Cleanup old entries	Yes (Admin)

Application Routes & Pages

Total Pages: 36

Authentication Pages

Located in `app/(auth)/` layout group:

Route	File	Description
/login	login/page.tsx	User login form
/register	register/page.tsx	User registration
/forgot-password	forgot-password/page.tsx	Password reset request
/reset-password	reset-password/page.tsx	Reset password with token
/auth/callback	auth/callback/page.tsx	SSO/OAuth callback handler

Admin Pages

Located in `app/admin/` :

Route	File	Description
/admin/companies	companies/page.tsx	Company management
/admin/companies/[id]	companies/[id]/page.tsx	Company detail page
/admin/exchanges	exchanges/page.tsx	Exchange management
/admin/traders	traders/page.tsx	Trader management

Main Application Pages

Located in `app/` :

Route	File	Description
/	page.tsx	Dashboard home
/aggregation	aggregation/page.tsx	Position aggregation

/alerts	alerts/page.tsx	Alert management
/api-docs	api-docs/page.tsx	API documentation (Swagger UI)
/api-keys	api-keys/page.tsx	API key management
/approvals	approvals/page.tsx	Approval workflows
/calendar	calendar/page.tsx	Trading calendar/schedules
/data-quality	data-quality/page.tsx	Data quality monitoring
/documents	documents/page.tsx	Document management
/exemptions	exemptions/page.tsx	Exemption management
/exemptions/request	exemptions/request/page.tsx	Request exemption
/exemptions/pending	exemptions/pending/page.tsx	Pending exemptions
/exemptions/history	exemptions/history/page.tsx	Exemption history
/filings	filings/page.tsx	Regulatory filings
/financial	financial/page.tsx	Financial management
/limits	limits/page.tsx	Position limits overview
/monitoring	monitoring/page.tsx	Real-time monitoring
/performance	performance/page.tsx	Performance analytics
/pre-trade	pre-trade/page.tsx	Pre-trade validation
/profile	profile/page.tsx	User profile
/reports	reports/page.tsx	Reports dashboard
/risk	risk/page.tsx	Risk management
/security	security/page.tsx	Security settings
/settings	settings/page.tsx	Application settings
/subscriptions	subscriptions/page.tsx	Subscription management
/support	support/page.tsx	Support tickets
/workflow-approvals	workflow-approvals/page.tsx	Workflow approvals

Error Pages

Route	File	Description
/error	error.tsx	Error boundary
/global-error	global-error.tsx	Global error handler

Services & Business Logic

Total Services: 35 (12,567 lines of code)

Authentication & Authorization Services

Service File	Lines	Purpose
auth-service.ts	450	JWT generation, password hashing, user authentication
oauth-service.ts	380	OAuth 2.0 integration (Google, Microsoft, etc.)
saml-service.ts	520	SAML SSO integration and IdP management
sso-provisioning-service.ts	280	Automatic user provisioning from SSO

Key Functions:

- generateToken() - Generate JWT access and refresh tokens
- hashPassword() - Bcrypt password hashing (10 rounds)
- verifyPassword() - Password verification
- validateSAMLAssertion() - SAML assertion validation
- provisionUser() - Auto-provision users from SSO attributes

Position & Risk Services

Service File	Lines	Purpose
limit-calculator.ts	22,981	Core position limit calculations (largest service)
pre-trade-service.ts	890	Pre-trade validation logic
pre-trade-validation-service.ts	1,240	CFTC-specific pre-trade checks
risk-analytics.ts	1,680	Risk metrics, VaR, stress testing

Key Functions:

- calculatePositionLimits() - Main limit calculation engine
- applySpreadNetting() - Apply spread netting rules
- calculateEquivalence() - Commodity equivalence calculations
- validateTrade() - Pre-trade validation
- calculateVaR() - Value at Risk calculation
- runStressTest() - Stress test scenarios

Compliance & Regulatory Services

Service File	Lines	Purpose
regulatory-compliance.ts	980	Compliance tracking and validation
regulatory-reporting.ts	1,450	CFTC, ICE, CME report generation
bid-week-service.ts	740	Bid week calendar and scheduling
exception-handling-service.ts	560	Exception workflow management

Key Functions:

- `generateCFTCReport()` - Generate CFTC LTRS report
- `generateICEReport()` - Generate ICE daily report
- `calculateBidWeek()` - Calculate bid week schedules
- `validateCompliance()` - Compliance rule validation

Data Management Services

Service File	Lines	Purpose
<code>data-quality-service.ts</code>	17,064	Data quality checks and validation (2nd largest)
<code>data-validation-service.ts</code>	620	Input validation and sanitization
<code>aggregation-service.ts</code>	1,890	Position aggregation logic
<code>batch-service.ts</code>	450	Batch processing and scheduling

Key Functions:

- `validateDataQuality()` - Run quality checks
- `reconcileData()` - Data reconciliation
- `aggregatePositions()` - Aggregate trader positions
- `processBatch()` - Batch transaction processing

Alerts & Notifications Services

Service File	Lines	Purpose
<code>alert-service.ts</code>	680	Alert generation and management
<code>email-service.ts</code>	31,545	Email notifications (largest service!)
<code>realtime-monitoring.ts</code>	920	Real-time position monitoring

Key Functions:

- `generateAlert()` - Create alert based on rules
- `sendEmail()` - Send email notifications
- `monitorPositions()` - Real-time monitoring cycle
- `detectBreach()` - Detect limit breaches

Documents & Support Services

Service File	Lines	Purpose
<code>document-service.ts</code>	540	Document management and versioning
<code>subscription-service.ts</code>	420	Subscription and billing
<code>subset-report-service.ts</code>	780	Subset report generation
<code>report-service.ts</code>	1,120	General report generation

Key Functions:

- `uploadDocument()` - Upload to R2 storage

- `versionDocument()` - Document versioning
- `generateReport()` - Report generation
- `exportToCSV()` - CSV export

Approvals & Workflow Services

Service File	Lines	Purpose
trade-approval-service.ts	480	Trade approval workflow

Key Functions:

- `submitForApproval()` - Submit trade for approval
- `approveRequest()` - Approve workflow request
- `rejectRequest()` - Reject workflow request

Security & Audit Services

Service File	Lines	Purpose
audit-logger.ts	320	Audit logging utility
audit-trail-service.ts	890	Audit trail management
security-logger.ts	280	Security event logging
api-key-service.ts	390	API key management

Key Functions:

- `logAudit()` - Log audit event
- `queryAuditTrail()` - Query audit history
- `logSecurityEvent()` - Log security event
- `generateAPIKey()` - Generate API key

Infrastructure Services

Service File	Lines	Purpose
cache-service.ts	240	KV caching layer
config-encryption-service.ts	180	Configuration encryption

Key Functions:

- `cacheGet()` - Get from KV cache
- `cacheSet()` - Set in KV cache with TTL
- `encryptConfig()` - Encrypt sensitive config
- `decryptConfig()` - Decrypt config

User Roles & Permissions

Role Hierarchy

Trade Nexus implements a 6-tier role-based access control system:

1. Super Admin (role_id: 1)

Highest Level - Platform Management

Capabilities:

- Manage all exchanges in the system
- Create and manage companies
- Full system access across all tenants
- Cross-company data access
- System configuration

Use Case: Platform administrators at Trade Nexus

Permissions: ALL (100+ permissions)

Data Scope: Global (all exchanges, all companies)

2. System Admin (role_id: 2)

System-Level Administration

Capabilities:

- System-level administration tasks
- User management across companies
- Cross-company data access
- System configuration
- Cannot manage exchanges

Use Case: IT administrators, system operators

Key Permissions:

- `system.configure`
- `users.create` , `users.update` , `users.delete`
- `companies.read` (all companies)
- `settings.update`
- `audit_trail.read`

Data Scope: Cross-company (within assigned scope)

3. Company Admin (role_id: 3)

Company-Level Management

Capabilities:

- Manage company users and traders
- Configure company-specific limits
- Assign traders to positions
- Company-level reporting
- User role management (within company)

Use Case: Company managers, operations managers

Key Permissions:

- `users.create` , `users.update` , `users.delete` (company-scoped)
- `traders.assign` , `traders.update`
- `market_limits.read` , `market_limits.update`
- `company.configure`
- `reports.generate`
- `exemptions.create`
- `alerts.configure`

Data Scope: Company-scoped (only their company data)

4. Trader (role_id: 4)

Trading Operations

Capabilities:

- Enter and manage own transactions
- View own positions and limits
- Submit trades for pre-trade validation
- Request exemptions
- View reports for own trading activity

Use Case: Traders, desk managers

Key Permissions:

- `transactions.create` , `transactions.read` (own data)
- `position_limits.read`
- `pre_trade.validate`
- `reports.read` (own data)
- `exemptions.request`

Data Scope: Trader-scoped (only own data)

Restrictions:

- Cannot view other traders' data
 - Cannot modify limits
 - Cannot approve exemptions
-

5. Compliance Officer (role_id: 5)

Compliance & Regulatory Oversight

Capabilities:

- Compliance oversight and monitoring
- Generate regulatory reports
- Manage exemptions (create, approve, deny)
- Access audit trails
- Configure alert rules
- Review and approve documents

Use Case: Compliance officers, risk managers

Key Permissions:

- exemptions.read , exemptions.create , exemptions.approve , exemptions.deny
- reports.generate , reports.schedule
- regulatory_filings.create , regulatory_filings.submit
- audit_trail.read
- alerts.read , alerts.configure
- documents.approve
- position_limits.read (all)
- transactions.read (all)

Data Scope: Company-scoped (full company visibility)

Special Access:

- Can view all trader data within company
- Cannot modify transactions or limits
- Focus on oversight and compliance

6. Auditor (role_id: 6)

Read-Only Audit Access

Capabilities:

- Read-only access to all data
- Audit trail review
- Report viewing
- No modification capabilities

Use Case: Internal auditors, external auditors, regulators

Key Permissions:

- audit_trail.read
- reports.read
- *.read (read access to all entities)
- NO write, update, or delete permissions

Data Scope: Depends on assignment (company-scoped or global)

Restrictions:

- Cannot create, update, or delete any data
- Cannot approve or reject requests
- Cannot configure system settings
- Pure observation and audit role

Permission System

Format: resource.action

Total Permissions: 100+

Example Permissions:

Market Limits:

- market_limits.read

- `market_limits.create`
- `market_limits.update`
- `market_limits.delete`
- `market_limits.override`

Transactions:

- `transactions.read`
- `transactions.create`
- `transactions.import`
- `transactions.delete`

Exemptions:

- `exemptions.read`
- `exemptions.create`
- `exemptions.update`
- `exemptions.delete`
- `exemptions.approve`
- `exemptions.deny`

Users:

- `users.read`
- `users.create`
- `users.update`
- `users.delete`
- `users.impersonate`

Reports:

- `reports.read`
- `reports.generate`
- `reports.export`
- `reports.schedule`

Alerts:

- `alerts.read`
- `alerts.create`
- `alerts.acknowledge`
- `alerts.configure`
- `alerts.delete`

And many more...

Multi-Tenancy & Data Scoping

Hierarchy:

```
Exchange (e.g., CFTC, ICE, CME)
├── Company (e.g., Trading Firm A)
│   └── Trader (e.g., John Doe, trader_code: JD001)
```

Data Isolation:

- All queries automatically scoped by `company_id`

- Implemented via `company-scoping` middleware
- Super Admin and System Admin can bypass scoping
- Traders can only see their own data (additional `trader_id` scoping)

Database Filtering:

- Every major table has `company_id` foreign key
 - Queries automatically include `WHERE company_id = ?`
 - Enforced at middleware level, not application level
 - Prevents accidental data leakage
-

Authentication & Security

Authentication Methods

1. Local Authentication (Email/Password)

Process:

1. User submits email and password
2. System validates email format
3. Queries `users` table for account
4. Verifies password with bcrypt
5. Checks if 2FA is enabled
6. If 2FA enabled, prompts for verification
7. Generates JWT access token (15 min expiry)
8. Generates JWT refresh token (7 day expiry)
9. Creates session record in database
10. Caches session in KV store
11. Returns tokens to client

Security Features:

- Bcrypt password hashing (10 rounds)
 - Password strength validation (OWASP guidelines)
 - Account lockout after 5 failed attempts
 - Password expiry (90 days)
 - Password history (prevent reuse of last 5 passwords)
-

2. SAML SSO

Supported:

- Company-specific SAML configurations
- Multiple IdPs per company
- Automatic user provisioning
- Role mapping from SAML attributes

Process:

1. User visits login page
2. System detects company domain
3. Initiates SAML authentication request
4. Redirects to company's IdP
5. User authenticates at IdP

6. IdP sends SAML assertion
7. System validates assertion signature
8. Extracts user attributes (email, name, role)
9. Provisions or updates user account
10. Maps SAML roles to Trade Nexus roles
11. Creates session and returns tokens

SAML Attributes Mapped:

- email → User email
 - firstName → First name
 - lastName → Last name
 - role → User role (mapped to role_id)
 - trader_code → Trader identifier
-

3. OAuth 2.0

Supported Providers:

- Google
- Microsoft
- Okta
- Auth0
- Custom OAuth providers

Process:

1. User clicks "Sign in with [Provider]"
 2. Redirects to OAuth provider
 3. User authorizes application
 4. Provider redirects with authorization code
 5. System exchanges code for access token
 6. Fetches user profile from provider
 7. Provisions or updates user account
 8. Creates session and returns tokens
-

Two-Factor Authentication (2FA)

Supported Methods:

1. TOTP (Time-based One-Time Password)

- Google Authenticator
- Authy
- Microsoft Authenticator
- Secret stored encrypted in user_2fa table

2. SMS Verification

- SMS code sent via Twilio/SNS
- 6-digit code
- 5-minute expiry

3. Email Verification

- Email code sent to registered email

- 6-digit code
- 10-minute expiry

4. Backup Codes

- 10 one-time use backup codes
- Securely hashed and stored
- Can be regenerated

5. Trusted Devices

- Device fingerprinting
- Trust device for 30 days
- Stored in `trusted_devices` table

Enforcement:

- Can be enforced at company level
- Optional or mandatory
- Configurable grace period for setup

Session Management

JWT Tokens:

- **Access Token:** Short-lived (15 minutes)
- **Refresh Token:** Long-lived (7 days)
- Both signed with RS256 (RSA signature)

Token Payload:

```
{
  "user_id": 123,
  "email": "user@example.com",
  "role_id": 3,
  "company_id": 5,
  "trader_code": "JD001",
  "permissions": [...],
  "iat": 1234567890,
  "exp": 1234568790
}
```

Session Storage:

- Database: `sessions` table (persistent)
- KV Store: Cached for fast validation
- TTL: Matches token expiry

Session Lifecycle:

1. Login creates session
2. Access token used for API requests
3. Token expires after 15 minutes
4. Client uses refresh token to get new access token
5. Refresh token can be used multiple times

6. Logout invalidates both tokens and session

Security:

- Token rotation on refresh
 - Refresh token family tracking (detect token theft)
 - Automatic revocation on suspicious activity
 - Session invalidation on password change
-

Security Middleware

1. Authentication Middleware (`auth.ts`)

- Validates JWT on every request
- Checks token signature and expiry
- Loads user permissions
- Attaches user context to request

2. Security Headers (`security-headers.ts`)

- Sets OWASP-recommended security headers
- Content Security Policy (CSP)
- X-Frame-Options: DENY
- X-Content-Type-Options: nosniff
- Strict-Transport-Security (HSTS)
- Referrer-Policy: no-referrer

3. Rate Limiter (`rate-limiter.ts`)

- Prevents brute force attacks
- IP-based rate limiting
- User-based rate limiting
- API key rate limiting

Rate Limit Presets:

- **Standard:** 100 requests/minute
- **Strict:** 20 requests/minute (auth endpoints)
- **Relaxed:** 1000 requests/minute (internal APIs)

4. Company Scoping (`company-scoping.ts`)

- Enforces multi-tenancy data isolation
- Automatically filters queries by `company_id`
- Prevents cross-company data access
- Bypassed only for Super Admin and System Admin

5. Performance Monitoring (`performance.ts`)

- Tracks request duration
- Logs slow queries (>1s)
- Records performance metrics

6. Error Tracking (`error-tracking.ts`)

- Captures and logs errors
- Integrates with Sentry

- Sanitizes sensitive data before logging
 - Tracks error rates and patterns
-

Additional Security Features

CORS Configuration:

- Origin whitelist
- Credentials allowed
- Preflight caching

SQL Injection Prevention:

- Parameterized queries only
- No string concatenation in queries
- ORM with prepared statements

XSS Protection:

- Input sanitization
- Output encoding
- CSP headers
- React's built-in XSS protection

CSRF Protection:

- CSRF tokens for state-changing operations
- SameSite cookie attribute
- Origin validation

API Key Authentication:

- Alternative to JWT for service-to-service
- Stored hashed in database
- Scoped permissions per key
- Usage tracking and rate limiting

Audit Logging:

- All authentication events logged
 - Failed login attempts tracked
 - Password changes logged
 - Permission changes logged
 - Session creation/termination logged
-

Database Schema

Schema Overview

Total Tables: 70+ **Total Migrations:** 27 **Database Engine:** SQLite (Cloudflare D1)

Core Table Categories

1. User & Authentication Tables

users

```

CREATE TABLE users (
  id INTEGER PRIMARY KEY AUTOINCREMENT,
  email VARCHAR(255) UNIQUE NOT NULL,
  password_hash TEXT NOT NULL,
  first_name VARCHAR(100),
  last_name VARCHAR(100),
  role_id INTEGER NOT NULL,
  company_id INTEGER,
  trader_code VARCHAR(50),
  status VARCHAR(20) DEFAULT 'active',
  email_verified BOOLEAN DEFAULT 0,
  last_login DATETIME,
  password_changed_at DATETIME,
  failed_login_attempts INTEGER DEFAULT 0,
  locked_until DATETIME,
  created_at DATETIME DEFAULT CURRENT_TIMESTAMP,
  updated_at DATETIME DEFAULT CURRENT_TIMESTAMP,
  FOREIGN KEY (role_id) REFERENCES roles(id),
  FOREIGN KEY (company_id) REFERENCES companies(id) ON DELETE CASCADE
);

```

roles

```

CREATE TABLE roles (
  id INTEGER PRIMARY KEY,
  name VARCHAR(50) UNIQUE NOT NULL,
  description TEXT,
  level INTEGER NOT NULL, -- Hierarchy level
  created_at DATETIME DEFAULT CURRENT_TIMESTAMP
);

```

permissions

```

CREATE TABLE permissions (
  id INTEGER PRIMARY KEY AUTOINCREMENT,
  resource VARCHAR(100) NOT NULL,
  action VARCHAR(50) NOT NULL,
  description TEXT,
  UNIQUE(resource, action)
);

```

role_permissions

```

CREATE TABLE role_permissions (
  role_id INTEGER NOT NULL,
  permission_id INTEGER NOT NULL,
  PRIMARY KEY (role_id, permission_id),
  FOREIGN KEY (role_id) REFERENCES roles(id) ON DELETE CASCADE,

```

```
    FOREIGN KEY (permission_id) REFERENCES permissions(id) ON DELETE CASCADE
);
```

sessions

```
CREATE TABLE sessions (
  id TEXT PRIMARY KEY,
  user_id INTEGER NOT NULL,
  access_token TEXT NOT NULL,
  refresh_token TEXT NOT NULL,
  expires_at DATETIME NOT NULL,
  ip_address VARCHAR(45),
  user_agent TEXT,
  created_at DATETIME DEFAULT CURRENT_TIMESTAMP,
  FOREIGN KEY (user_id) REFERENCES users(id) ON DELETE CASCADE
);
```

user_2fa

```
CREATE TABLE user_2fa (
  user_id INTEGER PRIMARY KEY,
  method VARCHAR(20) NOT NULL, -- 'totp', 'sms', 'email'
  secret TEXT, -- Encrypted TOTP secret
  phone VARCHAR(20), -- For SMS
  enabled BOOLEAN DEFAULT 0,
  backup_codes TEXT, -- JSON array of hashed codes
  created_at DATETIME DEFAULT CURRENT_TIMESTAMP,
  FOREIGN KEY (user_id) REFERENCES users(id) ON DELETE CASCADE
);
```

trusted_devices

```
CREATE TABLE trusted_devices (
  id INTEGER PRIMARY KEY AUTOINCREMENT,
  user_id INTEGER NOT NULL,
  device_fingerprint TEXT NOT NULL,
  device_name VARCHAR(255),
  last_used DATETIME,
  expires_at DATETIME,
  created_at DATETIME DEFAULT CURRENT_TIMESTAMP,
  FOREIGN KEY (user_id) REFERENCES users(id) ON DELETE CASCADE
);
```

2. Multi-Tenancy Tables

exchanges

```
CREATE TABLE exchanges (
  id INTEGER PRIMARY KEY AUTOINCREMENT,
```

```

name VARCHAR(255) NOT NULL,
code VARCHAR(50) UNIQUE NOT NULL, -- 'CFTC', 'ICE', 'CME'
description TEXT,
status VARCHAR(20) DEFAULT 'active',
created_at DATETIME DEFAULT CURRENT_TIMESTAMP,
updated_at DATETIME DEFAULT CURRENT_TIMESTAMP
);

```

companies

```

CREATE TABLE companies (
  id INTEGER PRIMARY KEY AUTOINCREMENT,
  name VARCHAR(255) NOT NULL,
  code VARCHAR(50) UNIQUE,
  exchange_id INTEGER NOT NULL,
  status VARCHAR(20) DEFAULT 'active',
  subscription_tier VARCHAR(50),
  max_traders INTEGER,
  max_positions INTEGER,
  settings JSON, -- Company-specific settings
  created_at DATETIME DEFAULT CURRENT_TIMESTAMP,
  updated_at DATETIME DEFAULT CURRENT_TIMESTAMP,
  FOREIGN KEY (exchange_id) REFERENCES exchanges(id)
);

```

trader_assignments

```

CREATE TABLE trader_assignments (
  id INTEGER PRIMARY KEY AUTOINCREMENT,
  trader_id INTEGER NOT NULL, -- References users.id
  company_id INTEGER NOT NULL,
  is_primary BOOLEAN DEFAULT 1,
  assigned_at DATETIME DEFAULT CURRENT_TIMESTAMP,
  FOREIGN KEY (trader_id) REFERENCES users(id) ON DELETE CASCADE,
  FOREIGN KEY (company_id) REFERENCES companies(id) ON DELETE CASCADE,
  UNIQUE(trader_id, company_id)
);

```

company_auth_configs

```

CREATE TABLE company_auth_configs (
  id INTEGER PRIMARY KEY AUTOINCREMENT,
  company_id INTEGER NOT NULL,
  auth_type VARCHAR(20) NOT NULL, -- 'saml', 'oauth', 'local'
  config JSON NOT NULL, -- IdP metadata, OAuth config, etc.
  is_default BOOLEAN DEFAULT 0,
  created_at DATETIME DEFAULT CURRENT_TIMESTAMP,
  updated_at DATETIME DEFAULT CURRENT_TIMESTAMP,
);

```

```
FOREIGN KEY (company_id) REFERENCES companies(id) ON DELETE CASCADE
);
```

3. Position & Limit Tables

market_limits

```
CREATE TABLE market_limits (
  id INTEGER PRIMARY KEY AUTOINCREMENT,
  commodity_code VARCHAR(50) NOT NULL,
  commodity_name VARCHAR(255),
  exchange_id INTEGER NOT NULL,
  spot_limit DECIMAL(18, 4),
  single_month_limit DECIMAL(18, 4),
  all_months_limit DECIMAL(18, 4),
  effective_date DATE NOT NULL,
  expiration_date DATE,
  status VARCHAR(20) DEFAULT 'active',
  created_at DATETIME DEFAULT CURRENT_TIMESTAMP,
  updated_at DATETIME DEFAULT CURRENT_TIMESTAMP,
  FOREIGN KEY (exchange_id) REFERENCES exchanges(id),
  INDEX idx_commodity (commodity_code),
  INDEX idx_effective_date (effective_date)
);
```

position_limits

```
CREATE TABLE position_limits (
  id INTEGER PRIMARY KEY AUTOINCREMENT,
  company_id INTEGER NOT NULL,
  trader_id INTEGER,
  commodity_code VARCHAR(50) NOT NULL,
  limit_type VARCHAR(50) NOT NULL, -- 'spot', 'single_month', 'all_months'
  limit_value DECIMAL(18, 4) NOT NULL,
  effective_date DATE NOT NULL,
  expiration_date DATE,
  status VARCHAR(20) DEFAULT 'active',
  created_by INTEGER,
  created_at DATETIME DEFAULT CURRENT_TIMESTAMP,
  updated_at DATETIME DEFAULT CURRENT_TIMESTAMP,
  FOREIGN KEY (company_id) REFERENCES companies(id) ON DELETE CASCADE,
  FOREIGN KEY (trader_id) REFERENCES users(id) ON DELETE SET NULL,
  INDEX idx_company_commodity (company_id, commodity_code),
  INDEX idx_trader (trader_id)
);
```

transactions

```
CREATE TABLE transactions (
  id INTEGER PRIMARY KEY AUTOINCREMENT,
```

```

company_id INTEGER NOT NULL,
trader_id INTEGER NOT NULL,
commodity_code VARCHAR(50) NOT NULL,
contract_month VARCHAR(20), -- 'F25', 'G25', etc.
trade_type VARCHAR(20) NOT NULL, -- 'buy', 'sell'
quantity DECIMAL(18, 4) NOT NULL,
price DECIMAL(18, 4),
trade_date DATE NOT NULL,
settlement_date DATE,
counterparty VARCHAR(255),
transaction_ref VARCHAR(100),
status VARCHAR(20) DEFAULT 'pending',
created_by INTEGER,
created_at DATETIME DEFAULT CURRENT_TIMESTAMP,
updated_at DATETIME DEFAULT CURRENT_TIMESTAMP,
FOREIGN KEY (company_id) REFERENCES companies(id) ON DELETE CASCADE,
FOREIGN KEY (trader_id) REFERENCES users(id),
INDEX idx_company_date (company_id, trade_date),
INDEX idx_trader_date (trader_id, trade_date),
INDEX idx_commodity (commodity_code)
);

```

limit_calculations

```

CREATE TABLE limit_calculations (
  id INTEGER PRIMARY KEY AUTOINCREMENT,
  company_id INTEGER NOT NULL,
  trader_id INTEGER,
  commodity_code VARCHAR(50) NOT NULL,
  calculation_date DATE NOT NULL,
  net_long_position DECIMAL(18, 4) DEFAULT 0,
  net_short_position DECIMAL(18, 4) DEFAULT 0,
  net_position DECIMAL(18, 4) DEFAULT 0,
  spot_position DECIMAL(18, 4) DEFAULT 0,
  single_month_max DECIMAL(18, 4) DEFAULT 0,
  all_months_position DECIMAL(18, 4) DEFAULT 0,
  applicable_limit DECIMAL(18, 4),
  limit_utilized_pct DECIMAL(5, 2),
  is_breach BOOLEAN DEFAULT 0,
  breach_amount DECIMAL(18, 4),
  calculated_at DATETIME DEFAULT CURRENT_TIMESTAMP,
  FOREIGN KEY (company_id) REFERENCES companies(id) ON DELETE CASCADE,
  FOREIGN KEY (trader_id) REFERENCES users(id) ON DELETE CASCADE,
  INDEX idx_company_commodity_date (company_id, commodity_code, calculation_date),
  INDEX idx_breaches (is_breach, calculation_date)
);

```

4. Risk Management Tables

risk_thresholds


```
CREATE TABLE risk_thresholds (
  id INTEGER PRIMARY KEY AUTOINCREMENT,
  company_id INTEGER NOT NULL,
  metric_name VARCHAR(100) NOT NULL,
  warning_level DECIMAL(18, 4),
  breach_level DECIMAL(18, 4),
  critical_level DECIMAL(18, 4),
  status VARCHAR(20) DEFAULT 'active',
  created_at DATETIME DEFAULT CURRENT_TIMESTAMP,
  updated_at DATETIME DEFAULT CURRENT_TIMESTAMP,
  FOREIGN KEY (company_id) REFERENCES companies(id) ON DELETE CASCADE
);
```

pre_trade_validations

```
CREATE TABLE pre_trade_validations (
  id INTEGER PRIMARY KEY AUTOINCREMENT,
  company_id INTEGER NOT NULL,
  trader_id INTEGER NOT NULL,
  commodity_code VARCHAR(50) NOT NULL,
  proposed_quantity DECIMAL(18, 4) NOT NULL,
  trade_type VARCHAR(20) NOT NULL,
  validation_result VARCHAR(20) NOT NULL, -- 'pass', 'warning', 'reject'
  violations JSON, -- Array of violation details
  simulated_position DECIMAL(18, 4),
  simulated_utilization DECIMAL(5, 2),
  validated_at DATETIME DEFAULT CURRENT_TIMESTAMP,
  FOREIGN KEY (company_id) REFERENCES companies(id),
  FOREIGN KEY (trader_id) REFERENCES users(id),
  INDEX idx_company_trader (company_id, trader_id),
  INDEX idx_validated_at (validated_at)
);
```

risk_scenarios

```
CREATE TABLE risk_scenarios (
  id INTEGER PRIMARY KEY AUTOINCREMENT,
  company_id INTEGER NOT NULL,
  name VARCHAR(255) NOT NULL,
  description TEXT,
  scenario_type VARCHAR(50), -- 'stress_test', 'monte_carlo', etc.
  parameters JSON,
  status VARCHAR(20) DEFAULT 'active',
  created_by INTEGER,
  created_at DATETIME DEFAULT CURRENT_TIMESTAMP,
  FOREIGN KEY (company_id) REFERENCES companies(id) ON DELETE CASCADE
);
```

5. Compliance Tables

limit_exemptions

```
CREATE TABLE limit_exemptions (  
  id INTEGER PRIMARY KEY AUTOINCREMENT,  
  company_id INTEGER NOT NULL,  
  trader_id INTEGER,  
  commodity_code VARCHAR(50) NOT NULL,  
  exemption_type VARCHAR(50) NOT NULL, -- 'hedge', 'spread', 'spot_month'  
  requested_limit DECIMAL(18, 4),  
  justification TEXT NOT NULL,  
  supporting_docs JSON, -- Array of document IDs  
  status VARCHAR(20) DEFAULT 'pending', -- 'pending', 'approved', 'denied'  
  requested_by INTEGER NOT NULL,  
  reviewed_by INTEGER,  
  reviewed_at DATETIME,  
  review_notes TEXT,  
  effective_date DATE,  
  expiration_date DATE,  
  created_at DATETIME DEFAULT CURRENT_TIMESTAMP,  
  FOREIGN KEY (company_id) REFERENCES companies(id) ON DELETE CASCADE,  
  FOREIGN KEY (trader_id) REFERENCES users(id),  
  FOREIGN KEY (requested_by) REFERENCES users(id),  
  FOREIGN KEY (reviewed_by) REFERENCES users(id),  
  INDEX idx_status (status),  
  INDEX idx_company (company_id)  
);
```

regulatory_filings

```
CREATE TABLE regulatory_filings (  
  id INTEGER PRIMARY KEY AUTOINCREMENT,  
  company_id INTEGER NOT NULL,  
  filing_type VARCHAR(50) NOT NULL, -- 'CFTC_LTRS', 'ICE_DAILY', 'CME'  
  report_date DATE NOT NULL,  
  filing_period VARCHAR(50), -- 'daily', 'weekly', 'monthly'  
  report_data JSON NOT NULL, -- Filing content  
  file_path VARCHAR(500), -- R2 storage path  
  status VARCHAR(20) DEFAULT 'draft', -- 'draft', 'submitted', 'accepted'  
  submitted_at DATETIME,  
  submitted_by INTEGER,  
  created_at DATETIME DEFAULT CURRENT_TIMESTAMP,  
  FOREIGN KEY (company_id) REFERENCES companies(id) ON DELETE CASCADE,  
  FOREIGN KEY (submitted_by) REFERENCES users(id),  
  INDEX idx_company_date (company_id, report_date),  
  INDEX idx_filing_type (filing_type)  
);
```

6. Alerts & Monitoring Tables

alerts

```

CREATE TABLE alerts (
  id INTEGER PRIMARY KEY AUTOINCREMENT,
  company_id INTEGER NOT NULL,
  user_id INTEGER, -- NULL for company-wide alerts
  alert_type VARCHAR(50) NOT NULL, -- 'breach', 'warning', 'info'
  severity VARCHAR(20) NOT NULL, -- 'critical', 'high', 'medium', 'low'
  title VARCHAR(255) NOT NULL,
  message TEXT NOT NULL,
  entity_type VARCHAR(50), -- 'position', 'limit', 'transaction'
  entity_id INTEGER,
  metadata JSON,
  status VARCHAR(20) DEFAULT 'unread', -- 'unread', 'read', 'acknowledged'
  acknowledged_by INTEGER,
  acknowledged_at DATETIME,
  created_at DATETIME DEFAULT CURRENT_TIMESTAMP,
  FOREIGN KEY (company_id) REFERENCES companies(id) ON DELETE CASCADE,
  FOREIGN KEY (user_id) REFERENCES users(id) ON DELETE CASCADE,
  FOREIGN KEY (acknowledged_by) REFERENCES users(id),
  INDEX idx_company_user (company_id, user_id),
  INDEX idx_status (status),
  INDEX idx_created (created_at)
);

```

position_breach_events

```

CREATE TABLE position_breach_events (
  id INTEGER PRIMARY KEY AUTOINCREMENT,
  company_id INTEGER NOT NULL,
  trader_id INTEGER,
  commodity_code VARCHAR(50) NOT NULL,
  breach_type VARCHAR(50) NOT NULL, -- 'spot', 'single_month', 'all_months'
  breach_date DATE NOT NULL,
  position_value DECIMAL(18, 4),
  limit_value DECIMAL(18, 4),
  breach_amount DECIMAL(18, 4),
  breach_percentage DECIMAL(5, 2),
  status VARCHAR(20) DEFAULT 'open', -- 'open', 'resolved', 'escalated'
  resolved_at DATETIME,
  resolution_notes TEXT,
  created_at DATETIME DEFAULT CURRENT_TIMESTAMP,
  FOREIGN KEY (company_id) REFERENCES companies(id) ON DELETE CASCADE,
  FOREIGN KEY (trader_id) REFERENCES users(id),
  INDEX idx_company_date (company_id, breach_date),
  INDEX idx_status (status)
);

```

7. Audit & Security Tables

audit_trail

```

CREATE TABLE audit_trail (
  id INTEGER PRIMARY KEY AUTOINCREMENT,
  user_id INTEGER,
  company_id INTEGER,
  action VARCHAR(100) NOT NULL, -- 'create', 'update', 'delete', 'login'
  entity_type VARCHAR(50), -- 'user', 'transaction', 'limit', etc.
  entity_id INTEGER,
  changes JSON, -- Before/after values
  ip_address VARCHAR(45),
  user_agent TEXT,
  timestamp DATETIME DEFAULT CURRENT_TIMESTAMP,
  FOREIGN KEY (user_id) REFERENCES users(id) ON DELETE SET NULL,
  FOREIGN KEY (company_id) REFERENCES companies(id) ON DELETE SET NULL,
  INDEX idx_user (user_id),
  INDEX idx_company (company_id),
  INDEX idx_entity (entity_type, entity_id),
  INDEX idx_timestamp (timestamp)
);

```

security_events

```

CREATE TABLE security_events (
  id INTEGER PRIMARY KEY AUTOINCREMENT,
  event_type VARCHAR(50) NOT NULL, -- 'failed_login', 'account_locked', etc.
  severity VARCHAR(20) NOT NULL,
  user_id INTEGER,
  ip_address VARCHAR(45),
  details JSON,
  timestamp DATETIME DEFAULT CURRENT_TIMESTAMP,
  FOREIGN KEY (user_id) REFERENCES users(id) ON DELETE SET NULL,
  INDEX idx_event_type (event_type),
  INDEX idx_timestamp (timestamp)
);

```

api_keys

```

CREATE TABLE api_keys (
  id INTEGER PRIMARY KEY AUTOINCREMENT,
  company_id INTEGER NOT NULL,
  key_name VARCHAR(255) NOT NULL,
  key_hash TEXT NOT NULL UNIQUE,
  key_prefix VARCHAR(20), -- First few chars for identification
  permissions JSON, -- Array of permission strings
  rate_limit INTEGER DEFAULT 100,
  status VARCHAR(20) DEFAULT 'active',
  last_used DATETIME,
  expires_at DATETIME,
  created_by INTEGER,
  created_at DATETIME DEFAULT CURRENT_TIMESTAMP,
  FOREIGN KEY (company_id) REFERENCES companies(id) ON DELETE CASCADE,

```

```
FOREIGN KEY (created_by) REFERENCES users(id),
INDEX idx_key_hash (key_hash),
INDEX idx_company (company_id)
);
```

Database Indexes

50+ Indexes for query optimization:

- Company scoping indexes (company_id)
- Date range indexes (trade_date, calculation_date)
- Status field indexes
- Foreign key indexes
- Composite indexes for common query patterns

Database Triggers

Auto-update triggers:

- Update updated_at timestamps
- Calculate document versions
- Log approval workflow changes
- Update bank balances on transactions

Database Views

Pre-built views for common queries:

```
CREATE VIEW v_active_subscriptions AS
SELECT c.id, c.name, cs.plan_name, cs.status, cs.expires_at
FROM companies c
JOIN company_subscriptions cs ON c.id = cs.company_id
WHERE cs.status = 'active';
```

```
CREATE VIEW v_open_tickets AS
SELECT st.*, u.email AS user_email, c.name AS company_name
FROM support_tickets st
JOIN users u ON st.user_id = u.id
JOIN companies c ON u.company_id = c.id
WHERE st.status IN ('open', 'in_progress');
```

```
CREATE VIEW v_pending_approvals AS
SELECT ar.*, u.email AS requester_email
FROM approval_requests ar
JOIN users u ON ar.requested_by = u.id
WHERE ar.status = 'pending';
```

Data Flow & Processing

Position Monitoring Data Flow

This diagram illustrates the complete data flow from transaction input through aggregation, limit calculation, monitoring, and regulatory compliance.

1. Data Input

Sources:

- **Trader Entry:** Manual transaction entry via web UI
- **CSV Import:** Bulk transaction upload
- **API Integration:** Programmatic transaction submission

Input Validation:

- Format validation (data types, required fields)
 - Business rule validation (trade dates, quantities)
 - Referential integrity checks
-

2. Data Validation & Quality

Format Validation:

- Field type checking
- Range validation
- Required field verification

Business Rules:

- Valid commodity codes
- Valid trade dates (not future, not holidays)
- Quantity > 0
- Valid counterparty

Quality Checks:

- Duplicate detection
- Anomaly detection (unusual quantities, prices)
- Consistency checks

Reconciliation:

- Match against external sources
 - Verify totals and balances
 - Identify discrepancies
-

3. Transaction Storage

Primary Storage:

- `transactions` table - permanent storage
- `temp_transactions` table - staging for aggregation

Metadata Captured:

- Transaction details
 - Created by, created at
 - Company and trader associations
 - Status tracking
-

4. Position Aggregation

Process Steps:

1. Group Transactions

- Group by company, trader, commodity
- Group by contract month
- Separate long and short positions

2. Apply Equivalence

- Convert related commodities to equivalent units
- Example: Convert physical gas to financial contracts
- Uses `position_equivalences` table

3. Spread Netting

- Net opposite positions in related months
- Reduce gross position to net position
- Apply spread netting rules

4. Calculate Net Position

- Net long vs short
- Calculate spot position (front month)
- Calculate single month max
- Calculate all months position

Result: Aggregated position by commodity and time period

5. Limit Calculation Engine

Steps:

1. Fetch Market Limits

- Query `market_limits` for commodity
- Get spot, single month, all months limits
- Check effective dates

2. Apply Exemptions

- Check for approved exemptions
- Modify limits if exemption exists
- Log exemption usage

3. Compute Utilization

- Calculate: $(\text{Net Position} / \text{Limit}) * 100$
- Determine breach status
- Calculate breach amount if over limit

4. Store Results

- Save to `limit_calculations` table
 - Store historical series in `limit_calculation_series`
 - Update `calculated_at` timestamp
-

6. Pre-Trade Validation (Alternative Path)

Process:

1. Simulate Position

- Add proposed trade to current position
- Calculate hypothetical new position

2. Check Limits

- Compare simulated position to limits
- Identify potential breaches

3. Validate Rules

- Check company-specific rules
- Check regulatory requirements
- Validate against risk thresholds

4. Return Result

- **Pass:** Trade allowed
- **Warning:** Trade allowed with notification
- **Reject:** Trade would breach limit

Use Case: Real-time validation before trade execution

7. Real-Time Monitoring

Monitoring Cycle:

1. Scan Positions

- Query `limit_calculations` for all active positions
- Run on schedule (every 15 minutes) or manual trigger

2. Detect Breaches

- Identify positions over limit
- Calculate breach severity
- Determine escalation needed

3. Generate Alerts

- Create alert records
- Determine recipients (traders, compliance, admins)
- Set alert severity

4. Log Events

- Create `position_breach_events` record
 - Log to audit trail
 - Send notifications
-

8. Regulatory Compliance

Compliance Workflow:

1. Check Bid Week

- Determine current bid week period
- Check if reporting is required
- Identify spot month

2. Generate Reports

- **CFTC LTRS:** Large Trader Reporting System
- **ICE Daily:** ICE exchange daily report
- **CME:** CME exchange report

3. File Reports

- Prepare report data (JSON)
- Generate PDF/XML file
- Store in R2 storage
- Update `regulatory_filings` table

4. Submit

- Submit to exchange/regulator
 - Track submission status
 - Store confirmation
-

9. Output & Reporting

Outputs:

Dashboard:

- Real-time position display
- Utilization charts and heatmaps
- Breach notifications
- Trending analysis

Reports:

- Position reports (by trader, commodity, time)
- Compliance reports
- Breach reports
- Historical reports

Regulatory Filings:

- CFTC, ICE, CME reports
- Downloadable PDF/XML
- Submission tracking

Notifications:

- Email alerts
 - In-app notifications
 - SMS for critical breaches
-

10. Audit Trail

Comprehensive Logging:

- All data inputs logged
- All calculations logged
- All alerts logged
- All compliance actions logged

Stored in:

- `audit_trail` table
- Includes: user, action, timestamp, changes, metadata

Use Cases:

- Compliance audit
- Regulatory inquiries
- Internal review
- Forensic analysis

Frontend Components

UI Component Library

Trade Nexus uses **shadcn/ui**, a collection of re-usable components built with Radix UI and Tailwind CSS.

Component Count: 20+ base UI components

Base UI Components

Located in `components/ui/` :

Component	Purpose
<code>alert.tsx</code>	Alert messages and notifications
<code>badge.tsx</code>	Status badges and labels
<code>button.tsx</code>	Buttons with variants
<code>card.tsx</code>	Card containers
<code>dialog.tsx</code>	Modal dialogs
<code>dropdown-menu.tsx</code>	Dropdown menus
<code>input.tsx</code>	Form input fields
<code>label.tsx</code>	Form labels
<code>separator.tsx</code>	Visual separators
<code>skeleton.tsx</code>	Loading skeletons
<code>tabs.tsx</code>	Tab navigation
<code>tooltip.tsx</code>	Tooltips
<code>select.tsx</code>	Select dropdowns

checkbox.tsx	Checkboxes
radio-group.tsx	Radio buttons
switch.tsx	Toggle switches
textarea.tsx	Multi-line text inputs
table.tsx	Data tables
toast.tsx	Toast notifications

Feature Components

Dashboard Components

Located in components/dashboard/ :

Component	Purpose
StatsCards.tsx	Summary statistics cards
PositionLimitsTable.tsx	Position limits table with sorting
AlertsTicker.tsx	Scrolling alert ticker
dashboard-charts.tsx	Charts and visualizations (Recharts)
dashboard-header.tsx	Dashboard header with filters
ice-positions.tsx	ICE position display
position-card.tsx	Individual position card
quick-actions.tsx	Quick action buttons
risk-card.tsx	Risk summary card
search-filters.tsx	Search and filter controls
top-risks-table.tsx	Top risks table

CFTC Components

Located in components/cftc/ :

Component	Purpose
audit-trail-viewer.tsx	Audit trail display and search
bid-week-status.tsx	Bid week status indicator
exceptions-dashboard.tsx	Exceptions management dashboard
pre-trade-validator.tsx	Pre-trade validation UI
subset-reports.tsx	Subset report viewer

Authentication Components

Located in components/auth/ :

Component	Purpose
auth-guard.tsx	Route protection and role checking
login-form.tsx	Login form with SSO options
register-form.tsx	Registration form
2fa-setup.tsx	Two-factor authentication setup

Layout Components

Located in components/layout/ :

Component	Purpose
DashboardLayout.tsx	Main dashboard layout with sidebar
main-layout.tsx	Application wrapper layout
sidebar.tsx	Navigation sidebar
header.tsx	Top navigation header
footer.tsx	Footer

Alert Components

Located in components/alerts/ :

Component	Purpose
alerts-bell.tsx	Notification bell with count
alert-list.tsx	List of alerts
alert-detail.tsx	Alert detail view

Theme Components

Located in components/theme/ :

Component	Purpose
theme-initializer.tsx	Theme initialization on load
theme-toggle.tsx	Dark/light mode toggle
theme-provider.tsx	Theme context provider

Shared Components

Component	Location	Purpose
error-boundary.tsx	components/	Error boundary wrapper
news-ticker.tsx	components/	News ticker
loading-spinner.tsx	components/	Loading spinner
breadcrumbs.tsx	components/	Breadcrumb navigation
pagination.tsx	components/	Pagination controls

State Management

Zustand Stores

auth-store.ts (8,732 lines)

- User authentication state
- Login/logout actions
- Token management
- Permission checking

alerts-store.ts

- Alert notifications
- Unread count
- Mark as read/acknowledged

dashboard-store.ts

- Dashboard data caching
- Filter state
- Date range selection

theme-store.ts

- Dark/light mode preference
- Theme persistence

Styling

Tailwind CSS Configuration:

- Custom color palette
- Custom spacing scale
- Responsive breakpoints
- Custom animations

Dark Mode Support:

- Automatic theme detection
- Manual theme toggle
- Persistent preference
- CSS variable based theming

Deployment & Infrastructure

Cloudflare Platform

Services Used:

1. Cloudflare Workers

- Backend API runtime
- Edge computing
- 0ms cold starts
- Automatic scaling

2. Cloudflare Pages

- Frontend hosting
- Automatic deployments from Git
- Preview deployments for PRs
- Global CDN distribution

3. Cloudflare D1

- SQLite database
- Edge-optimized queries
- Automatic backups
- Point-in-time recovery

4. Cloudflare KV

- Key-value store
- Session caching
- Low-latency reads
- Global replication

5. Cloudflare R2

- Object storage (S3-compatible)
- Document storage
- Report archival
- No egress fees

Deployment Process

Backend Deployment:

```
# Deploy backend to Workers
npm run deploy

# Deploy to specific environment
wrangler deploy --env production
```

Frontend Deployment:

```
# Build frontend
npm run build

# Deploy to Pages
wrangler pages deploy
```

Database Migrations:

```
# Apply migrations to D1
wrangler d1 migrations apply trade-nexus-db

# Rollback migration
wrangler d1 migrations rollback trade-nexus-db
```

Environment Configuration

Development:

- Local D1 database
- Local KV namespace
- Dev R2 bucket
- API endpoint: `http://localhost:8787`

Staging:

- Staging D1 database
- Staging KV namespace
- Staging R2 bucket
- API endpoint: `https://staging-api.tradenexus.com`

Production:

- Production D1 database
 - Production KV namespace
 - Production R2 bucket
 - API endpoint: `https://api.tradenexus.com`
-

Performance

Response Times:

- P50: <50ms
- P95: <200ms
- P99: <500ms

Availability:

- 99.99% uptime SLA
- Multi-region redundancy
- Automatic failover

Scalability:

- Auto-scaling based on demand

- No manual capacity planning
 - Handles 10,000+ req/sec
-

Monitoring

Cloudflare Analytics:

- Request volume
- Error rates
- Response times
- Geographic distribution

Sentry Integration:

- Error tracking
- Performance monitoring
- Release tracking
- User feedback

Custom Metrics:

- API endpoint usage
 - Feature adoption
 - User activity
 - Database query performance
-

Security

SSL/TLS:

- Automatic SSL certificates
- TLS 1.3 support
- HTTPS enforcement

DDoS Protection:

- Cloudflare's DDoS mitigation
- Rate limiting
- Bot protection

Firewall:

- Web Application Firewall (WAF)
- Custom firewall rules
- IP allowlisting/blocklisting

Data Encryption:

- Encryption at rest (D1, KV, R2)
 - Encryption in transit (TLS)
 - Encrypted session storage
-

Backup & Recovery

Database Backups:

- Automatic daily backups
- Point-in-time recovery (7 days)
- Manual backup triggers

Disaster Recovery:

- Multi-region data replication
- Automatic failover
- Recovery Time Objective (RTO): <1 hour
- Recovery Point Objective (RPO): <1 hour

Appendix

Technology Version Reference

Technology	Version	Latest Stable
Node.js	20.x	20.11.0
Next.js	15.1.3	15.1.3
React	18.3.1	18.3.1
TypeScript	5.3.3	5.3.3
Hono	4.6.14	4.6.14
Zustand	5.0.1	5.0.1
Tailwind CSS	3.4.0	3.4.0
Recharts	2.12.7	2.12.7

API Response Formats

Success Response:

```
{
  "success": true,
  "data": { ... },
  "meta": {
    "timestamp": "2025-10-29T12:00:00Z",
    "request_id": "abc123"
  }
}
```

Error Response:

```
{
  "success": false,
  "error": {
    "code": "VALIDATION_ERROR",
    "message": "Invalid input data",
  }
}
```

```
    "details": [
      {
        "field": "quantity",
        "message": "Must be greater than 0"
      }
    ],
    "meta": {
      "timestamp": "2025-10-29T12:00:00Z",
      "request_id": "abc123"
    }
  }
}
```

Rate Limits

Endpoint Category	Rate Limit	Window
Authentication	20 req/min	1 minute
Standard API	100 req/min	1 minute
Reporting	50 req/min	1 minute
Bulk Operations	10 req/min	1 minute
API Key	1000 req/min	1 minute